

PRODUCT REQUIREMENTS DOCUMENT

(PRD)

BOQ AI

Version: 2.0 MVP

Category: Construction Technology (ConTech)

Product Owner: Olushola Samuel Ariyo

Status: MVP Approved

1. PRODUCT OVERVIEW

BOQ AI is an Artificial Intelligence-powered construction cost estimation platform that converts Architectural PDF Drawings into professional Bills of Quantities (BOQs), Material Schedules, Labour Schedules, Construction Stage Cost Reports, and Preliminary Project Cost Estimates.

The platform is designed for:

- Homeowners
- Property Developers
- Contractors
- Architects
- Quantity Surveyors
- Construction Consultants

Users upload Architectural PDF Drawings and receive a complete construction cost breakdown within minutes.

Unlike traditional BOQ software, BOQ AI produces reports in plain language so that both technical and non-technical users can understand construction costs and project requirements.

2. PRODUCT VISION

To become the world's most trusted AI-powered construction estimating and cost intelligence platform.

3. PRODUCT MISSION

To make construction costing simple, transparent, and accessible for everyone.

4. PROBLEM STATEMENT

Preparing a Bill of Quantities currently requires:

- Manual measurements
- Professional quantity surveyors
- Significant time investment
- High consultancy fees

Many property owners begin projects without understanding:

- Total project cost
- Material requirements
- Labour requirements
- Construction stage costs

This often results in:

- Budget overruns
- Delayed projects
- Material shortages
- Cost disputes

BOQ AI solves these challenges through automated quantity takeoff and intelligent cost estimation.

5. TARGET USERS

Homeowners

Need to know:

- Total project cost
- Material quantities
- Cost by construction stage

Property Developers

Need:

- Feasibility studies
- Preliminary budgets
- Cost planning

Contractors

Need:

- Tender preparation
- Material schedules
- Labour schedules

Architects

Need:

- Cost feedback during design

Quantity Surveyors

Need:

- Automated quantity takeoff
- Faster BOQ production

6. CORE USER FLOW

Step 1

Create Account

Step 2

Verify Email

Step 3

Select Country

Step 4

Select State / Region

Step 5

Create Project

Step 6

Upload Architectural Drawing PDF

Step 7

AI Drawing Analysis

Step 8

Generate BOQ

Step 9

Review Reports

Step 10

Export PDF or Excel

7. AI PROCESSING ENGINE

After upload, AI shall:

Drawing Recognition

Identify:

- Floor Plans
- Elevations
- Sections
- Roof Plans
- Site Plans

Element Detection

Detect:

- Rooms
- Walls
- Doors
- Windows
- Columns
- Roof Areas
- Floor Areas
- Ceiling Areas

Quantity Extraction

Calculate:

- Floor Area
- Wall Area
- Roof Area
- Ceiling Area
- Window Count
- Door Count

8 CONSTRUCTION STAGE ENGINE.

BOQ AI shall organize project costs according to actual construction stages.

Stage 1

Preliminaries

Examples:

- Site clearing
- Mobilization
- Temporary works

Stage 2

Substructure

Examples:

- Excavation
- Foundation
- Hardcore
- Concrete
- Damp Proof Membrane

Stage 3

Superstructure

Examples:

- Blockwork
- Columns
- Beams
- Lintels

Stage 4

Roofing

Examples:

- Roof Structure
- Roofing Sheets

- Fascia
- Rainwater Goods

Stage 5

Doors & Windows

Examples:

- Security Doors
- Internal Doors
- Aluminum Windows

Stage 6

Finishes

Examples:

- Floor Finishes
- Wall Finishes
- Ceiling Finishes
- Painting

Stage 7

External Works

Examples:

- Drainage
- Walkways
- Fencing
- Landscaping

9. BILL OF QUANTITY GENERATION

The system shall generate professional BOQ sheets.

Columns:

- S/N
- Description
- Plain Language Explanation
- Unit
- Quantity
- Rate
- Amount

Example:

225mm Block Wall

Plain Language:

Cost of constructing external walls using 225mm blocks.

10. PROFESSIONAL REPORT OUTPUT

Every report shall contain:

Section 1

Project Information

- Project Name
- Location
- Drawing Uploaded
- Date Generated

Section 2

Executive Summary

Total Project Cost

Estimated Duration

Highest Cost Stage

Highest Cost Material

Top Cost Drivers

11. THREE-LAYER COST SUMMARY

Layer 1

Labour Cost Summary

Construction Stage Labour Cost

Foundation	₦X
Blockwork	₦X
Roofing	₦X
Finishes	₦X

Total Labour Cost

Layer 2

Material Quantity + Cost Summary

| Material | Quantity | Unit | Rate | Amount |

Examples:

- Cement
- Blocks
- Sand
- Granite
- Steel Reinforcement
- Paint
- Roofing Sheets

Total Material Cost

Layer 3

Material Quantity Summary (Without Cost)

| Material | Quantity | Unit |

Examples:

- 150 Bags Cement
- 4,000 Blocks
- 45 Tons Granite
- 35 Tons Sand

Purpose:

Material procurement planning.

12. STAGE-BY-STAGE COST REPORT

BOQ AI shall generate a dedicated report showing:

Stage	Cost
Preliminaries	N X
Substructure	N X
Superstructure	N X
Roofing	N X
Finishes	N X
External Works	N X

This helps users understand project cashflow requirements.

13. PLAIN LANGUAGE PROJECT EXPLANATION

The system shall generate a human-friendly explanation.

Example:

"This project is estimated to cost ₦28,500,000.

Materials account for ₦18,000,000 while labour accounts for ₦10,500,000.

The most expensive stage is roofing.

You will require approximately:

- 180 bags of cement
- 4,500 blocks
- 50 tons of granite

The estimated construction duration is 24 weeks."

14. REGIONAL PRICING ENGINE

Users select:

- Country
- State
- City

The system shall apply local market rates automatically.

Examples:

- Lagos
- Ibadan
- Abuja
- Port Harcourt
- Kano

15. PROJECT HISTORY

Store:

- Uploaded Drawings
- BOQs
- Cost Reports
- Material Schedules
- Generated Dates

Users can:

- Reopen
- Compare
- Duplicate
- Recalculate

Projects

16. EXPORTS

Export Formats:

- PDF
- Excel

Every export shall contain:

1. Executive Summary
2. Labour Summary
3. Material Cost Summary
4. Material Quantity Summary
5. Stage Cost Summary
6. Detailed BOQ
7. Plain Language Explanation
8. AI Disclaimer

17. SUBSCRIPTION PLANS

Free

- 2 Projects Monthly
- Watermarked Reports
- No Export

Professional

- Unlimited Projects
- Material Schedule
- Regional Pricing
- Excel Export

Enterprise

- Team Collaboration
- Company Dashboard
- API Access
- Custom Rate Libraries

18. PERFORMANCE REQUIREMENTS

- Upload Processing < 30 Seconds
- BOQ Generation < 5 Minutes
- 99.9% Uptime

19. SUCCESS METRICS

- BOQs Generated
- Monthly Active Users
- Subscription Revenue
- Export Count
- User Satisfaction Score

20. MVP LIMITATIONS

Version 2 MVP Supports:

- Architectural PDF Drawings Only

Version 3 Roadmap:

- Structural Drawings
- Electrical Drawings
- Plumbing Drawings
- Mechanical Drawings
- BIM Integration
- Construction Scheduling
- Procurement Management
- Contractor Marketplace

17. TECHNICAL REQUIREMENTS

17.1 System Architecture

BOQ AI shall use a modular AI-first architecture consisting of:

1. Frontend Application
2. Backend API Layer
3. AI Processing Engine
4. Drawing Recognition Engine
5. Quantity Extraction Engine
6. Cost Estimation Engine
7. Report Generation Engine
8. Database Layer

9. Cloud Storage Layer

The system must be designed to support future integration with:

- Structural Drawings
- Electrical Drawings
- Plumbing Drawings
- BIM Models
- Procurement Systems
- Contractor Marketplaces

17.2 Frontend Architecture

Technology Stack:

- Next.js 15+
- React 19+
- TypeScript
- Tailwind CSS
- ShadCN UI
- TanStack Query

Responsibilities:

- User Authentication
- Project Dashboard
- Drawing Upload
- BOQ Viewer
- Report Viewer
- Export Management
- Subscription Management

Requirements:

- Mobile Responsive
- Tablet Responsive
- Desktop Optimized
- Progressive Web App (PWA) Ready

17.3 Backend Architecture

Technology Stack:

- FastAPI
- Python
- REST API
- Async Processing

Responsibilities:

- Authentication
- File Processing
- AI Job Queue
- BOQ Generation
- Pricing Engine
- Report Generation
- Subscription Management

Requirements:

- Modular Service Architecture
- API Versioning
- Background Job Processing
- Horizontal Scalability

17.4 AI Processing Architecture

The AI Engine shall process uploaded drawings through multiple stages.

Stage 1

Document Understanding

Functions:

- PDF Parsing
- Drawing Classification
- Scale Detection
- Sheet Identification

Stage 2

Computer Vision Analysis

Functions:

- Wall Detection
- Room Detection
- Door Detection
- Window Detection
- Roof Detection
- Floor Area Detection

Recommended Technologies:

- OpenCV

Stage 3

Quantity Extraction

Functions:

- Area Calculations
- Length Calculations
- Count Calculations
- Material Estimation

Outputs:

- Quantities
- Dimensions
- Material Requirements

Stage 4

Construction Logic Engine

Converts extracted quantities into:

- Substructure BOQ
- Superstructure BOQ
- Roofing BOQ
- Finishes BOQ
- External Works BOQ

Stage 5

Cost Estimation Engine

Applies:

- Regional Rates
- Labour Rates
- Material Rates

Outputs:

- Cost Breakdown
- Stage Costs
- Total Project Cost

17.5 AI Model Requirements

Primary LLM:

- DeepSeek Models

The architecture shall support switching models without requiring system redesign.

Required Capabilities:

- Drawing Interpretation
- Quantity Validation
- Plain Language Explanation
- Report Generation

17.6 Database Requirements

Database:

PostgreSQL

Core Tables:

Users

Projects

Drawings

Generated BOQs

Material Schedules

Labour Schedules

Regional Rates

Subscriptions

Exports

Audit Logs

Requirements:

- ACID Compliance
- Daily Backups
- High Availability

17.7 Cloud Storage

Storage Service:

AWS S3

Store:

- Uploaded Drawings
- Generated Reports
- Exported Files
- User Assets

Requirements:

- Encrypted Storage
- Automatic Backups
- Version Control

17.8 Authentication & Authorization

Authentication Methods:

- Email & Password
- Google OAuth

Future Support:

- Microsoft Login
- Apple Login

Role Types:

1. Homeowner
2. Contractor
3. Architect
4. Quantity Surveyor
5. Enterprise Admin

Permissions shall be role-based.

17.9 Pricing Engine Architecture

The Pricing Engine shall support:

Country Level Pricing

State Level Pricing

City Level Pricing

Custom Company Pricing

Rate Categories:

- Labour Rates
- Material Rates
- Equipment Rates

The system shall allow rate updates without code deployment.

17.10 Construction Stage Engine

The Stage Engine shall organize all generated quantities into:

1. Preliminaries
2. Substructure
3. Superstructure

4. Roofing
5. Doors & Windows
6. Finishes
7. External Works

Each stage shall have:

- Material Quantities
- Material Costs
- Labour Costs
- Stage Total Cost

This enables project cashflow forecasting.

17.11 Report Generation Engine

The Report Engine shall generate:

Professional BOQ

Material Schedule

Labour Schedule

Construction Stage Report

Executive Summary

Plain Language Report

Supported Formats:

- PDF
- Excel

Future:

- Word
- CSV

17.12 Export Requirements

Every exported report must contain:

1. Project Information
2. Executive Summary
3. Labour Cost Summary
4. Material Quantity & Cost Summary
5. Material Quantity Summary
6. Construction Stage Cost Summary
7. Detailed BOQ
8. Plain Language Explanation
9. AI Disclaimer

17.13 Performance Requirements

Drawing Upload:

Less than 30 seconds

Drawing Analysis:

Less than 2 minutes

BOQ Generation:

Less than 5 minutes

Export Generation:

Less than 30 seconds

System Availability:

99.9%

17.14 Security Requirements

Data Encryption:

AES-256

Transport Security:

TLS 1.3

Requirements:

- Encrypted Storage
- Secure File Upload
- Rate Limiting
- DDOS Protection
- Role-Based Access Control
- Audit Logging

17.15 Monitoring & Analytics

Track:

- BOQs Generated
- Processing Time
- AI Accuracy
- Active Users
- Subscription Revenue
- Export Frequency

Tools:

- PostHog
- Sentry
- OpenTelemetry

17.16 Scalability Requirements

The platform shall support:

- 100,000+ Users
- 1,000,000+ Projects
- Multi-Country Deployment
- Multi-Currency Support

Future Expansion:

- BIM Integration
- Procurement Management
- Construction Scheduling
- Contractor Marketplace
- Mobile Applications

PRODUCT SUCCESS STATEMENT

A user with no construction experience should be able to upload an Architectural PDF Drawing, generate a professional BOQ, understand every construction stage, understand material requirements, understand labour requirements, understand project cashflow requirements, and obtain a reliable preliminary construction estimate within five minutes.